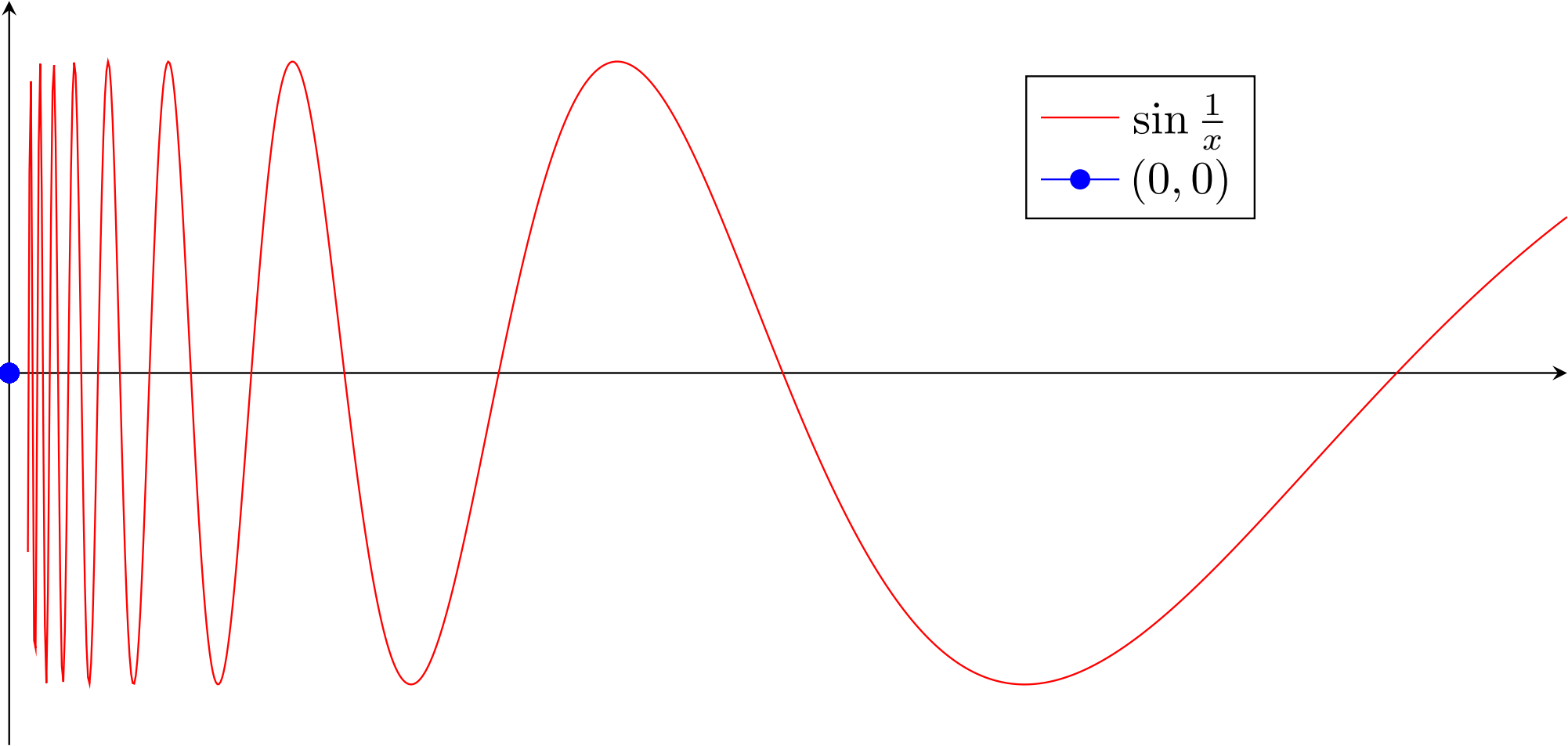
$\sin \frac{1}{x}$
$(0, 0)$